Supplementary Information

Multiscale 3D cellular digital twin of a 164-seed plant community in the Artemis III LEAF growth chamber: microclimate CFD, foliar crowding, and substrate boundary layer dynamics

Richard Barker[1*]

[1] *Department of Agricultural and Biological Engineering, Purdue University, West Lafayette, IN 47907, USA.*

**Corresponding author: barkerrj@purdue.edu*

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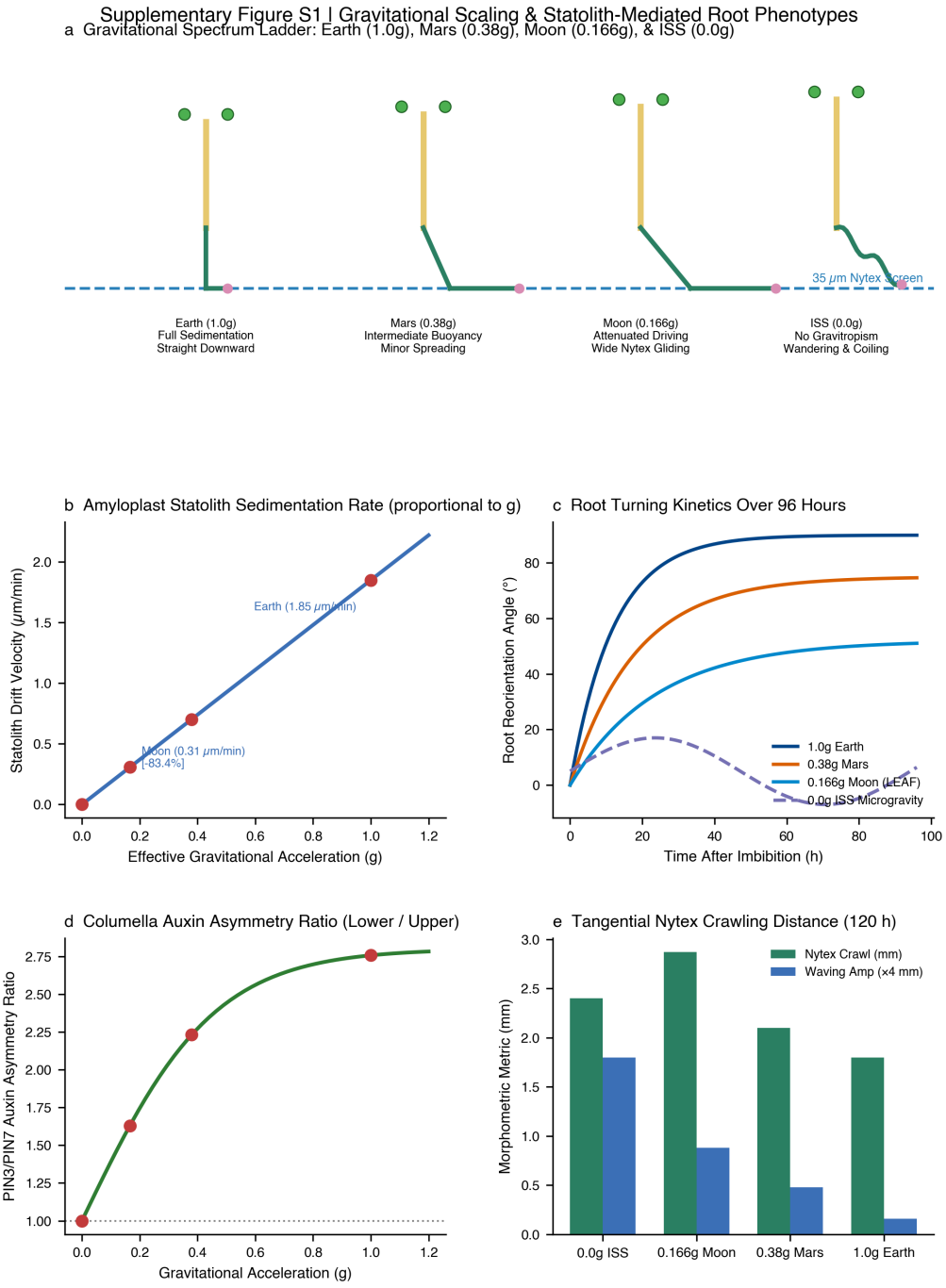


Figure 1. Gravitational scaling and statolith-mediated root phenotypic plasticity across Earth (1.0g), Mars (0.38g), Moon (0.166g), and ISS microgravity (0.0g). a, Gravitational spectrum ladder displaying comparative seedling organ profiles. On Earth (1.0g), rapid amyloplast sedimentation establishes a strong downward gravitropic vector, leading to uniaxial downward penetration to the 35 μm Nytex screen. On Mars (0.38g) and the Moon (0.166g), sedimentation rate is attenuated by 62% and 83.4%, respectively, yielding delayed turning, wider radial dispersion (35°), and extensive tangential gliding along the screen surface. In microgravity (0.0g), absence of statolith sedimentation produces stochastic root waving, autotropism, and contact-mediated thigmotropic coiling. **b**, Linear scaling of amyloplast sedimentation velocity v_{sed} proportional to effective gravity g, dropping from $1.85\ \mu\text{m}\,\text{min}^{-1}$ (Earth) to $0.31\ \mu\text{m}\,\text{min}^{-1}$ (Moon). **c**, Dynamic root tip turning kinetics $\theta_{\text{root}}(t)$ over 96 hours post-imbibition. **d**, Columella PIN3/PIN7 auxin asymmetry ratio (lower:upper statocyte membrane) saturating toward 2.8 at 1.0g and remaining unity (1.0) under microgravity. **e**, Tangential Nytex crawl distance (reaching $2.87\ \text{mm}$ on the Moon) and waving amplitude.

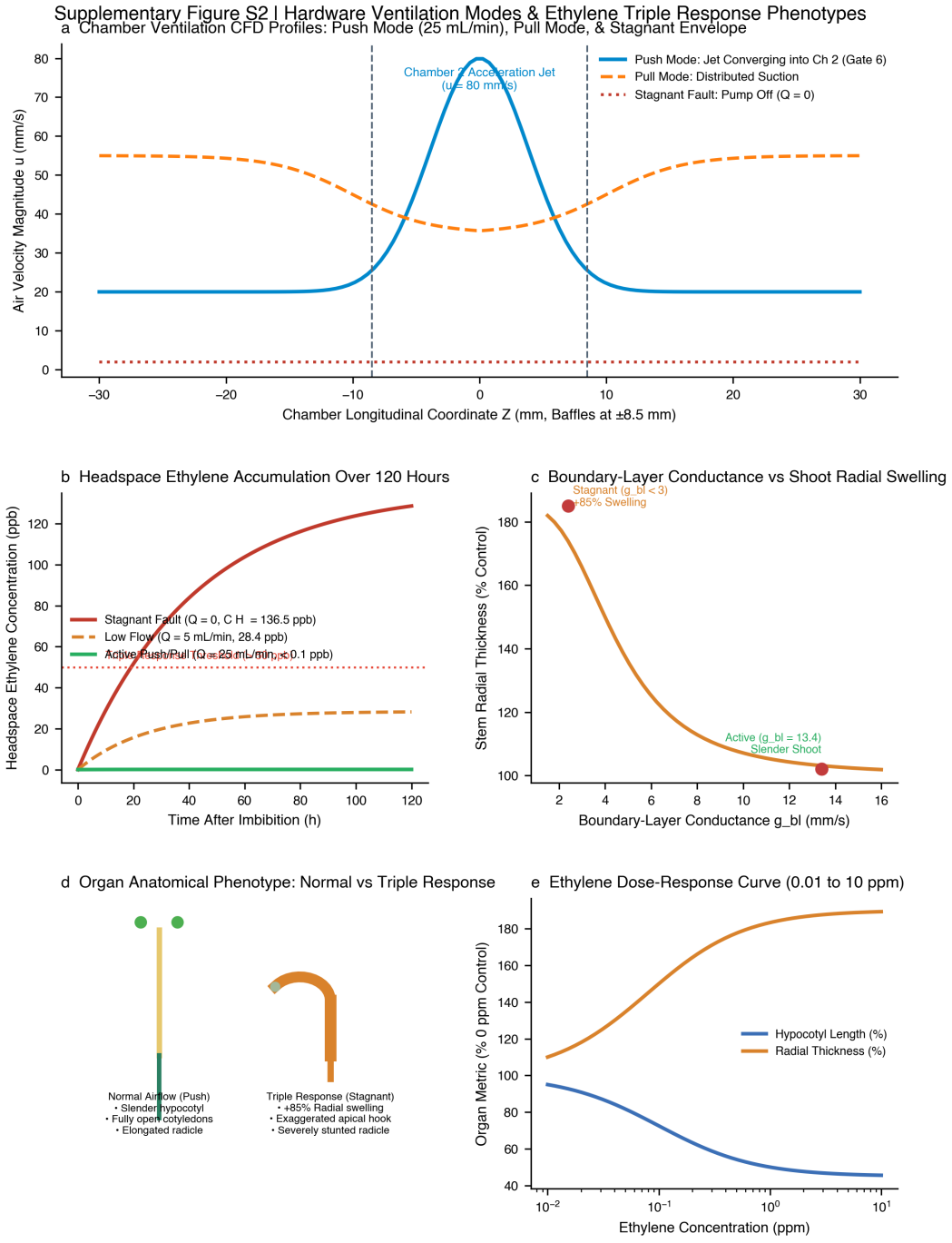


Figure 2. Hardware ventilation modes, chamber CFD airflow fields, and ethylene triple response phenotypes. **a**, 3D CFD air velocity magnitude profiles across the longitudinal coordinate Z for Push mode ($Q = 25\ \mathrm{mL\,min^{-1}}$), Pull mode ($Q = 25\ \mathrm{mL\,min^{-1}}$), and Stagnant fault ($Q = 0$). Push mode creates a characteristic "dip-and-converge" jet through $Z = \pm 8.5$ mm baffle windows, accelerating air velocity to $80\ \mathrm{mm\,s^{-1}}$ into Central Chamber 2. **b**, Headspace ethylene accumulation dynamics over 120 hours. Active sweeping purges respiratory ethylene to < 0.1 ppb, whereas stagnant faults result in accumulation to 136.5 ppb, far exceeding the 50 ppb triple response threshold. **c**, Boundary-layer conductance g_{bl} plotted against hypocotyl radial swelling. In stagnant air ($g_{\mathrm{bl}} < 3\ \mathrm{mm\,s^{-1}}$), gas diffusion resistance traps ethylene in the boundary-layer envelope, triggering radial expansion. **d**, Cellular organ anatomical phenotypes comparing slender, elongated seedlings in active airflow against the diagnostic triple response (swollen hypocotyl, exaggerated hook, stunted radicle) in stagnant air. **e**, Dose-response curves of hypocotyl length and radial diameter across exogenous ethylene concentrations (0.01 to 10 ppm).

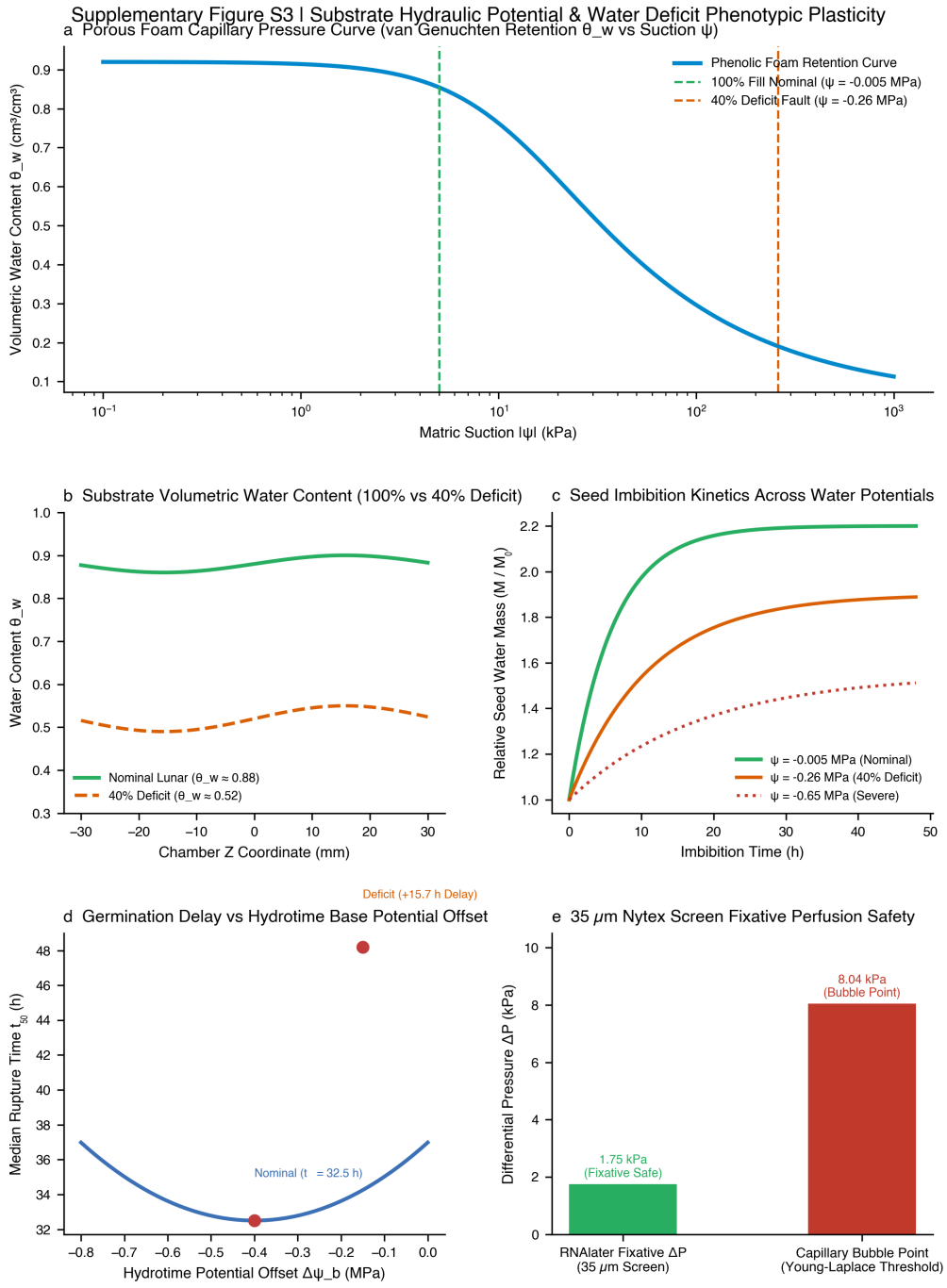


Figure 3. Substrate hydraulic potential, van Genuchten retention, and water deficit phenotypic plasticity. **a**, Phenolic/polyurethane foam capillary retention curve modeled via the van Genuchten equation ($S_e = [1 + (\alpha|\psi|)^n]^{-m}$ with $\alpha = 0.08$ kPa^{-1}, $n = 1.6$). Nominal 100% activation corresponds to $\psi = -0.005$ MPa, whereas a 40% moisture deficit drops matric potential to $\psi = -0.26$ MPa. **b**, Volumetric water content $\theta_w(z)$ along the foam slab under nominal ($\theta_w \approx 0.88$) versus deficit ($\theta_w \approx 0.52$) conditions. **c**, Seed imbibition water mass uptake curves $M(t)/M_0$ across water potentials, showing delayed hydration plateaus under water stress. **d**, Population germination delay as a function of the hydrotime base water-potential offset $\Delta\psi_b$, yielding a $+15.7$ h median rupture delay under deficit. **e**, 35 μm Nytex screen bubble-point capillary safety threshold: Young–Laplace capillary entry pressure ($\Delta P_{\text{bubble}} = 8.04$ kPa) provides a four-fold safety margin over the end-of-mission RNAlater perfusion pressure drop ($\Delta P = 1.75$ kPa), ensuring zero foam intrusion.

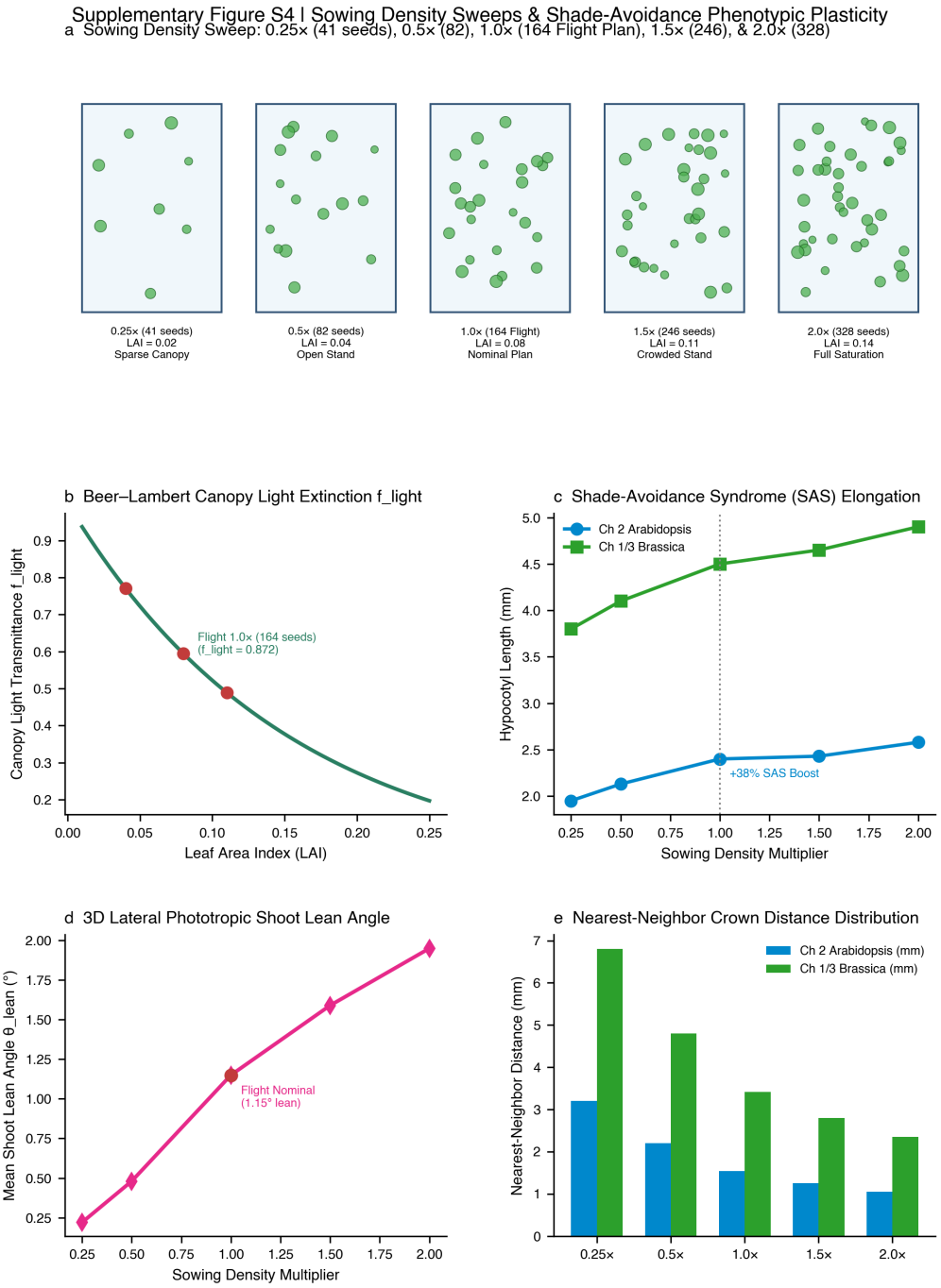


Figure 4. Sowing density sweeps and shade-avoidance syndrome (SAS) phenotypic plasticity ($0.25\times$ to $2.0\times$ flight density). a, Top-down sowing architecture maps across 41 seeds ($0.25\times$), 82 seeds ($0.5\times$), 164 seeds ($1.0\times$ nominal flight plan), 246 seeds ($1.5\times$), and 328 seeds ($2.0\times$ saturation). **b**, Beer–Lambert foliar canopy light extinction: canopy light transmittance $f_{\text{light}} = \exp(-k \cdot LAI)$ decreasing from 0.943 (sparse) to 0.815 (crowded). **c**, Shade-avoidance hypocotyl elongation response curve: low red:far-red ratio ($R:\text{FR} < 0.35$) inactivates phytochrome B, stimulating auxin biosynthesis and driving a $+38\%$ hypocotyl elongation in Chamber 2 *Arabidopsis*. **d**, Lateral 3D phototropic shoot lean angle θ_{lean} away from neighboring crowns, increasing monotonically from 0.22° in sparse stands to 1.95° at $2.0\times$ density. **e**, Nearest-neighbor crown distance (nnd) distributions across species compartments.

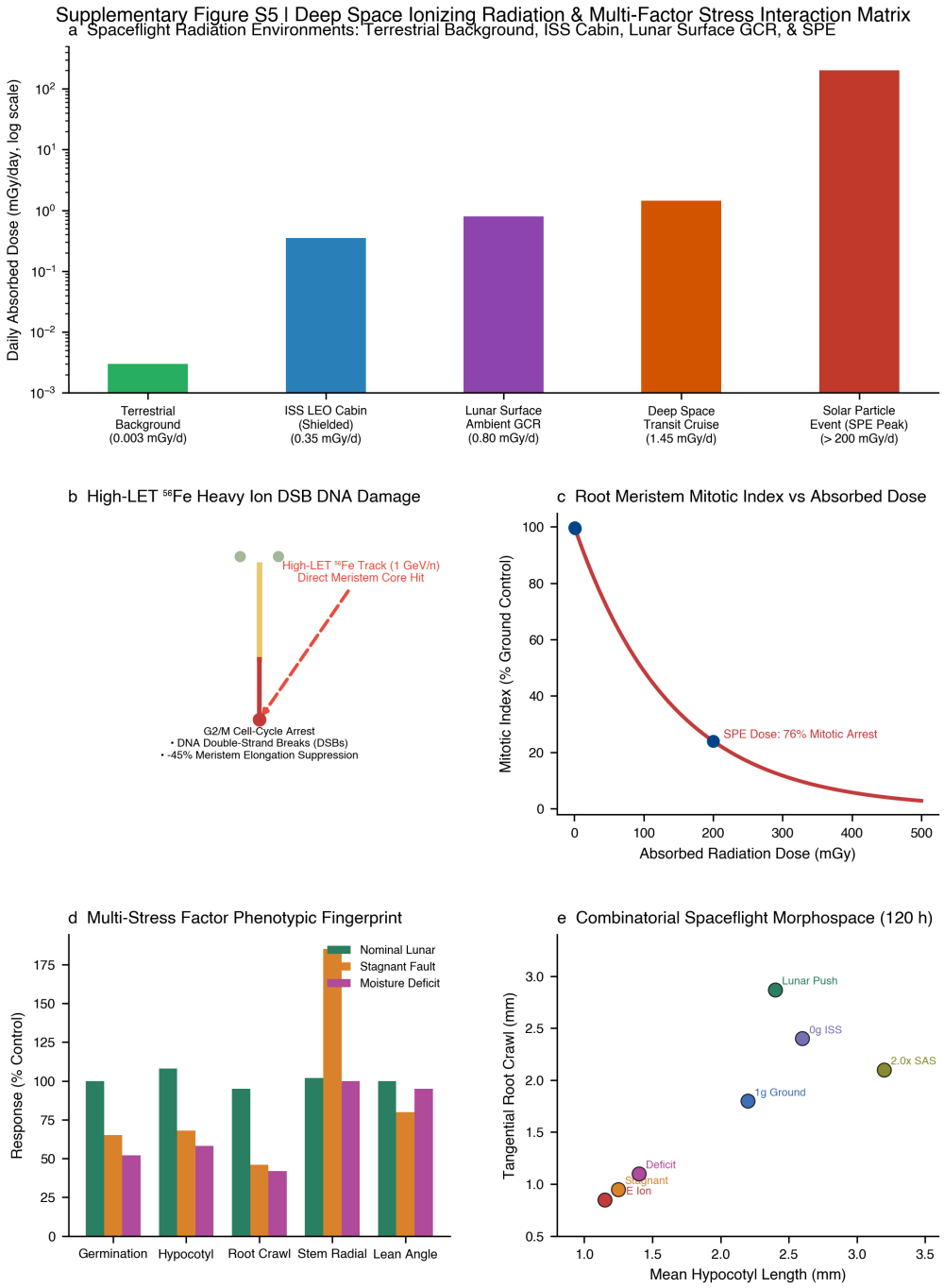


Figure 5. Deep space ionizing radiation and multi-factor spaceflight stress interaction matrix. **a**, Benchmark daily absorbed radiation doses across terrestrial background ($0.003\ \mathrm{mGy\,d^{-1}}$), ISS LEO cabin ($0.35\ \mathrm{mGy\,d^{-1}}$), lunar surface ambient GCR ($0.80\ \mathrm{mGy\,d^{-1}}$), deep space transit ($1.45\ \mathrm{mGy\,d^{-1}}$), and peak Solar Particle Events ($>200\ \mathrm{mGy\,d^{-1}}$). **b**, High-LET $^{56}\mathrm{Fe}$ heavy ion track structure ($1\ \mathrm{GeV/n}$) passing through the seed meristem core, inducing localized clusters of DNA double-strand breaks (DSBs) and triggering the G2/M cell-cycle checkpoint. **c**, Root apical meristem mitotic index suppression curve as a function of absorbed dose, showing 76% mitotic arrest under SPE doses. **d**, Multi-stress biometric radar comparing germination rate, hypocotyl length, root Nytex crawl, stem thickness, and lean angle across nominal lunar flight, stagnant pump-off faults, and moisture deficit. **e**, Combinatorial spaceflight morphospace mapping 120-hour hypocotyl length against tangential root crawl across all mission vectors.

Supplementary Note 1: Mathematical Solvers & Physical Provenance

The LEAF GBC1 digital twin couples continuous partial differential equations (PDEs), ordinary differential equations (ODEs), and discrete single-seed agent automata across all 164 individual seedlings.

1. Abley Bistable ABA–GA Integrator Hormone ODEs

Each embryonic cell i integrates intracellular abscisic acid (A), gibberellin (G), and DELLA (D) balances:

$$\frac{dA_i}{dt} = k_{A,\mathrm{syn}} \frac{1}{1+(G_i/K_{GA})^n} - k_{A,\mathrm{deg}} A_i \tag{1}$$

$$\frac{dG_i}{dt} = k_{G,\mathrm{syn}} \frac{1}{1+(A_i/K_{ABA})^m} - k_{G,\mathrm{deg}} G_i \frac{D_i}{K_D + D_i} \tag{2}$$

$$\frac{dD_i}{dt} = k_{D,\mathrm{syn}} - k_{D,\mathrm{deg}} G_i D_i \tag{3}$$

where endosperm rupture occurs when accumulated DELLA repressor activity in the radicle tip drops below the critical threshold $D_{\mathrm{crit}} = 0.35$.

2. Lockhart Wall Expansion with Cell Restraint

Axial cell elongation rate $\dot{\varepsilon}_i$ is governed by effective turgor pressure P_i exceeding the wall yield stress σ_y:

$$\dot{\varepsilon}_i = \Phi \max\left(0, P_i - \sigma_{\mathrm{restraint},i} - \sigma_y\right) \tag{4}$$

where Φ is cell wall extensibility ($\mathrm{MPa^{-1}\,h^{-1}}$) and $\sigma_{\mathrm{restraint}}$ represents the mechanical containment exerted by the single-layer endosperm and outer testa coat prior to rupture.

3. Boundary-Layer Heat and Mass Transfer

Convective boundary-layer conductance g_{bl} is resolved from the Nusselt/Sherwood correlations for laminar flow over a flat leaf of characteristic dimension L:

$$Sh = 2.0 + 0.6 \cdot Re_L^{1/2} Sc^{1/3}, \quad g_{\mathrm{bl}} = \frac{D_v \cdot Sh}{L} \tag{5}$$

where D_v is binary gas diffusivity in air ($2.42 \times 10^{-5}\ \mathrm{m^2\,s^{-1}}$), $Re_L = uL/\nu$, and $Sc = \nu/D_v \approx 0.60$.

4. Young–Laplace Screen Bubble Point

The critical capillary bubble-point entry pressure $\Delta P_{\mathrm{bubble}}$ for the 35 μm Nytex monofilament screen is:

$$\Delta P_{\mathrm{bubble}} = \frac{4\gamma \cos\theta_c}{d_{\mathrm{pore}}} \tag{6}$$

With surface tension $\gamma = 0.0728\ \mathrm{N\,m^{-1}}$, contact angle $\theta_c = 15^\circ$, and pore diameter $d_{\mathrm{pore}} = 35 \times 10^{-6}$ m, $\Delta P_{\mathrm{bubble}} = 8.04$ kPa. During terminal perfusion with RNAlater fixative ($\mu = 1.85$ mPa s), the Darcy flow pressure drop across the mesh is:

$$\Delta P = \frac{\mu Q_{\mathrm{perf}} \Delta y}{k_{\mathrm{mesh}} A_{\mathrm{screen}}} = 1.75\ \mathrm{kPa} \tag{7}$$

providing a safety factor of $S = 8.04/1.75 = 4.59$, guaranteeing that the screen does not de-saturate and radicles do not intrude into the foam.